

Advanced Machine Learning

DATA 642

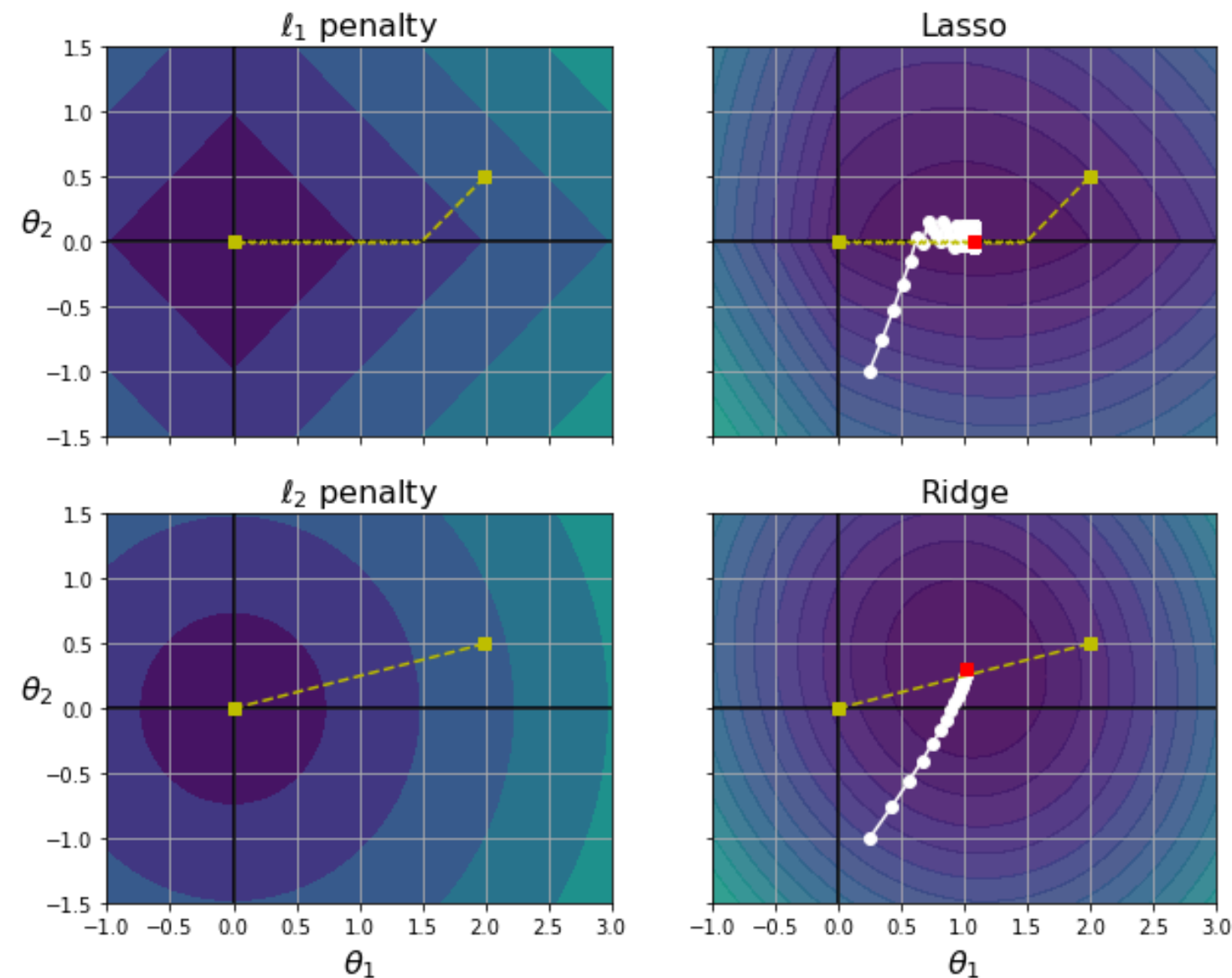
Lecture 4 — Optimization

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, and Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier.

From an optimization perspective

On the left plot, the contours represent an unregularized MSE cost function plus an ℓ_1 penalty with $\lambda = 0.5$. The global minimum is on the $\theta_2 = 0$ axis. In this case, the gradient descent path first reaches $\theta_2 = 0$ and then reaches the global minimum.

On the right plot, the the contours represent an unregularized MSE cost function plus an ℓ_2 penalty.



Elastic Net

Elastic Net is a regularization technique that combines both ℓ_1 (Lasso) and ℓ_2 (Ridge) penalties in the linear regression model.

Combines advantages of Lasso and Ridge:

1. Lasso: Promotes sparsity, feature selection.
2. Ridge: Handles multicollinearity, stabilizes coefficient estimates.

The Elastic Net objective function is defined as:

$$J(\boldsymbol{\theta}) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda_1 \|\boldsymbol{\theta}\|_1 + \lambda_2 \|\boldsymbol{\theta}\|_2^2 \quad (1)$$

Advantages

- Effective for high-dimensional data with multicollinearity.
- Allows feature selection while handling correlated predictors.
- Provides a compromise between Lasso and Ridge regularization.

Disadvantages

- Complexity in tuning parameters.
- Less intuitive interpretation compared to individual methods.
- Computational cost can be higher.
- Less sparse solutions compared to Lasso.
- Sensitivity to correlation structure among features.